

- 4 Leaves that have fallen from trees are an important source of nutrients for many organisms living in streams.

Bacteria and fungi can break down molecules such as cellulose and lignin, releasing nutrients for small animals that live in the water.

Decomposition of two types of leaf in the Mahoning River, Ohio, was investigated.

- (a) (i) Which row of the table describes the structure of cellulose?

(1)

	Type of glucose molecule	Type of glycosidic bond
☐ A	alpha (α)	1–4
☐ B	alpha (α)	1–6
☐ C	beta (β)	1–4
☐ D	beta (β)	1–6

- (ii) Cellulose and lignin are broken down by enzymes released from the bacteria and fungi.

Fungi belong to the domain Eukarya.

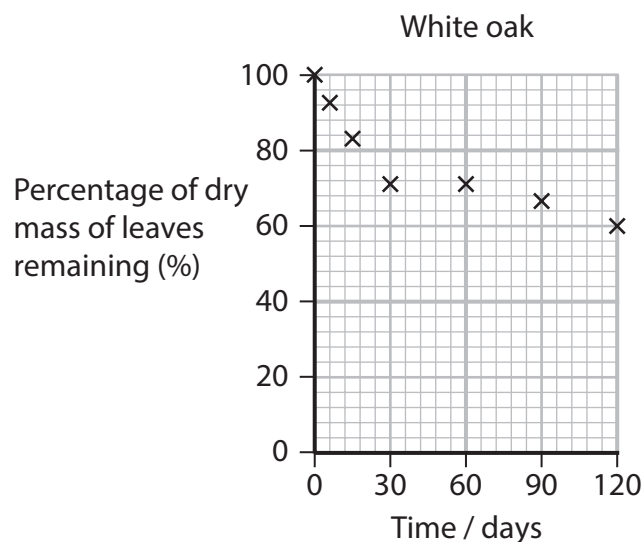
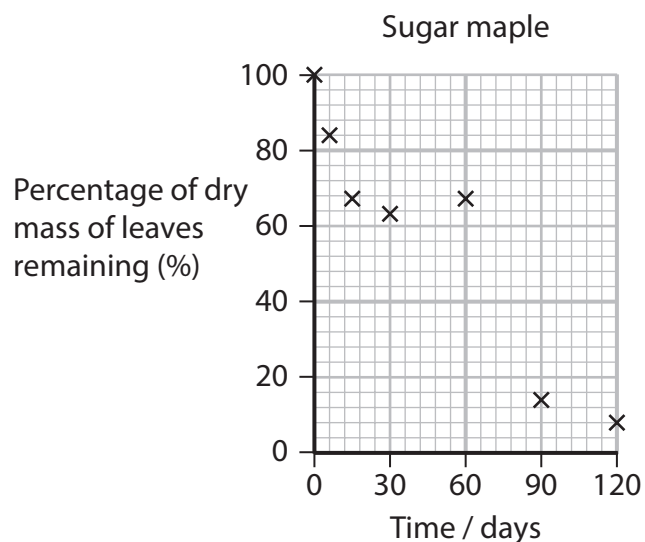
Which of the following statements are correct for **both** bacteria and fungi?

- 1 The enzymes are synthesised on ribosomes
- 2 The genes coding for these enzymes are found in the nucleus
- 3 Enzymes are released by endocytosis

(1)

- ☐ A 1 only
- ☐ B 1 and 2 only
- ☐ C 2 only
- ☐ D 2 and 3 only

(b) The graphs show the extent to which two types of leaf decomposed in 120 days.



(i) Decomposition was measured as a decrease in dry mass of the leaves.

Suggest why dry mass was measured and not wet mass.

(1)

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(ii) Calculate the rate of decrease in percentage dry mass between 30 days and 120 days for the sugar maple leaves.

Give your answer to **two** significant figures.

(2)

Answer

(iii) Describe **two** conclusions that can be made about the decomposition of sugar maple leaves compared with the white oak leaves.

(2)

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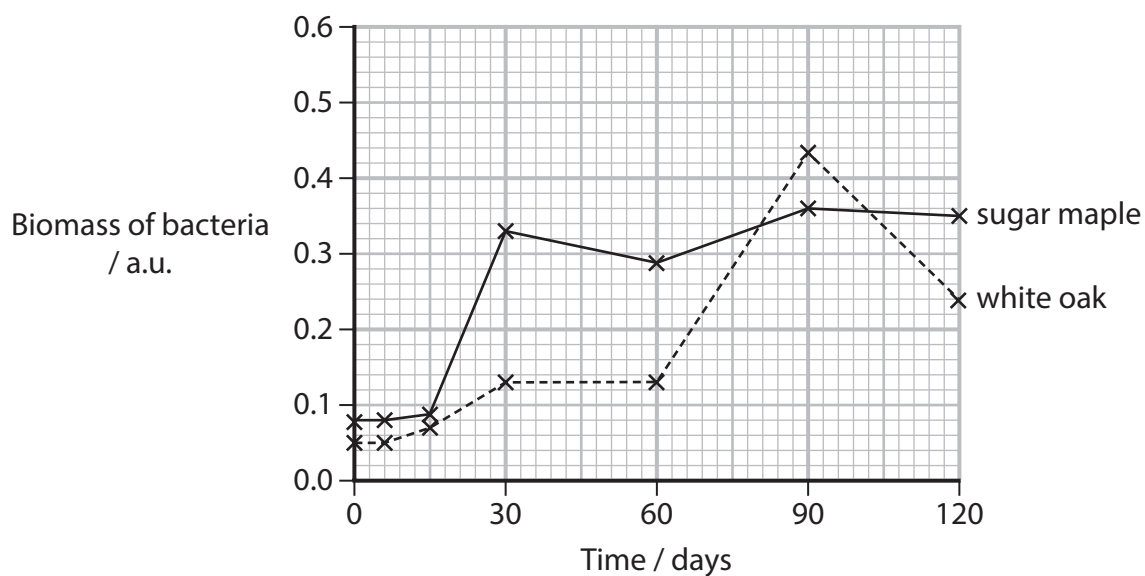
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(c) The biomass of bacteria on the two types of leaf during the 120-day period was determined.

The graph shows the changes in biomass of bacteria on these leaves.



(i) Calculate how many times greater the biomass of bacteria on the sugar maple leaves is after 30 days compared with the biomass of bacteria at 0 days.

(1)

Answer

- (ii) Suggest why the increase in biomass of bacteria after 30 days is greater on the sugar maple leaves than it is on the white oak leaves.

Give a reason for your answer.

(2)

- (iii) Suggest why the biomass of bacteria on the white oak leaves decreased after 90 days.

Give a reason for your answer.

(2)

(Total for Question 4 = 12 marks)